Care as Compass: A Formal Theory of Design Form Grounded in Morphospace, Selection Landscape, and Multi-Scale Agency

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Abstract Design theory evaluates proposals on criteria such as logic, correctness, clarity, and originality, yet these criteria lack grounding in a shared formal framework. No existing theory simultaneously provides a syntactic account of form, an epistemic account of the design process, and an explanation of why certain forms arise rather than others. Three formal traditions each address part of the problem but remain disconnected: Stiny's shape algebra provides syntax, Hatchuel and Weil's C-K theory provides epistemic logic, and Levin's multi-scale competence framework provides dynamics. We propose FORMLÆRE, a formal design theory structured as seven propositions that unify these traditions into a single generative equation: Form = $SG(\nabla_{C(A)} L(c,t))$. The theory introduces *care*; the dimensionality of the agent's cognitive cone; as the measure of design intelligence. We test the theory against a corpus of 2,066 European chairs (1280–2024) from Nasjonalmuseet and the Victoria and Albert Museum, supplemented by 2,204 three-dimensional mesh models. Twenty statistical analyses across 14 cross-validation subsets yield results consistent with the theory's core predictions: style predicts 1.9–2.3× more form variation than material, historian-defined style categories do not form discrete morphological clusters (silhouette = -0.338), and the occupied morphospace expands cumulatively (553× hull growth). We demonstrate the theory's explanatory reach through five worked examples spanning the steel tube chair race of 1925–1928, Neri Oxman's Silk Pavilion, AI-assisted design tools, conversational agent design, and continental-scale ecological restoration. The theory yields seven immediate, falsifiable research programs.

Keywords design theory, morphospace, fitness landscape, shape grammar, C-K theory, multi-scale agency, care, selection pressure

1. Introduction

In 1859, Michael Thonet patented chair No. 14: six steam-bent beechwood components, joined by ten screws and two nuts. The design was so efficient that 36 disassembled chairs fit in a single cubic meter for shipping. By 1930, over fifty million had been produced. The chair remains in continuous production today. Why this form?

The standard answer; "form follows function" (Sullivan, 1896); fails on inspection. The function of a chair is to afford sitting to a human body. But this affordance is shared by every chair ever made: the Windsor, the Wassily, the Eames Lounge, the monobloc plastic garden chair. Sitting is constant across the class. It explains why chairs exist but carries zero information about why one chair takes one form rather than another (Michl, 1995). If function determined form, all chairs would be identical. They are not.

This paper proposes a formal theory that answers the question. We call it FORMLÆRE (Norwegian: *theory of form*). The theory is structured as seven propositions that build from foundational definitions to an ethical conclusion:

1. The world of form is all that is the case.
2. Not all positions are equally probable.
3. The selection pressures produce a landscape.
4. The landscape is dynamic.
5. Agents exist.
6. Form arises between agents.
7. Form shows how far care reached.

The theory synthesizes three formal traditions that have developed independently: George Stiny's shape algebra and form grammars (Stiny, 2006; Stiny & Gips, 1972), which provide the *syntax* of form; Armand Hatchuel and Benoit Weil's C-K theory (Hatchuel & Weil, 2003, 2009), which provides the *epistemic logic* of the design process; and Michael Levin's multi-scale competence framework (Doctor et al., 2024; Fields & Levin, 2022; Levin, 2022), which provides the *dynamics* of agent-driven form production. The synthesis produces a single generative equation:

$$\text{Form} = SG\big(\nabla_{C(A)}\, L(c,\,t)\big)$$

Form is the grammar's response to the landscape gradient as perceived through the agent's cognitive cone. The theory introduces *care*; the dimensionality of the agent's representational capacity; as the measure of design intelligence.

1.1. The problem: ungrounded evaluation criteria

Design theory currently evaluates theoretical proposals on criteria such as "logic," "correctness," "clarity," and "originality." These criteria, while reasonable in isolation, lack grounding in a shared formal framework. What counts as "correct" in design theory? Correct relative to what formal structure? What counts as "original" if there is no shared map of the theoretical landscape?

The consequence is that design theory operates without consensus on its own foundations. Shape grammars describe the syntax of form but do not explain why forms arise. C-K theory describes the logic of design reasoning but provides no morphological structure. Fitness landscape models from evolutionary biology explain pattern formation but have not been formalized for designed artifacts. And recent work on multi-scale agency and care (Doctor et al., 2024) provides a compelling account of intelligence as dimensional attention but has not been connected to design practice.

1.2. Contribution

This paper makes three contributions:

1. **Theoretical.** We present a formal design theory that unifies shape algebra, C-K theory, and multi-scale competence into a single framework. The theory is expressed as seven falsifiable propositions with a core generative equation.
2. **Empirical.** We test the theory against a corpus of 2,066 European chairs (1280–2024) drawn from two museum collections, with 2,204 supplementary three-dimensional mesh models. Twenty statistical analyses across 84 cross-validation tests confirm the theory's core predictions; no postulate is falsified.
3. **Practical.** We demonstrate the theory's explanatory reach through five worked examples spanning historical design (the cantilever chair race of 1925–1928), bio-fabrication (Neri Oxman's Silk Pavilion), AI-assisted design (Figma Weave), conversational agent design, and continental-scale ecological restoration (the Great Green Wall). Each example is described in concrete factual detail, not as a hypothetical illustration.

The paper proceeds as follows. Section 2 reviews the three source traditions and identifies the synthesis gap. Section 3 presents the seven propositions with formal definitions and the core equation. Section 4 tests the theory empirically. Section 5 applies the theory to five worked examples. Section 6 discusses explanatory power and limitations. Section 7 proposes seven falsifiable research programs. Section 8 concludes.

2. Theoretical Background

The theory synthesizes three formal traditions that have developed independently. Each addresses a necessary aspect of design form; syntax, epistemic logic, and dynamics; but none provides a complete account. This section reviews each tradition and identifies the gap the synthesis fills.

2.1. Shape algebra and form grammars: the syntax of form

George Stiny and James Gips introduced shape grammars in 1972 as a generative specification for visual design (Stiny & Gips, 1972). A shape grammar is a rule system that operates on forms through embedding: a rule recognizes a subshape within the current form and replaces it with another. The key insight is that rules fire on the *present geometry*, not on construction history. A form has no memory of how it was made; the grammar sees only what is there now.

Stiny subsequently formalized the underlying algebra of shapes (Stiny, 1991, 2006). Forms are elements of a Boolean algebra closed under sum (union), difference, and product (intersection, derived from sum and difference). The transformation group of Euclidean space provides rotation, translation, and scaling. The embedding relation $\tau(a) \leq s$ determines whether a subshape a can be found within a shape s under some transformation τ.

Shape grammars have been applied to architectural analysis (Stiny & Gips, 1972), Palladian villas, Chinese ice-ray lattices, Hepplewhite chair backs, and computational design synthesis. The formalism provides a precise syntactic account: it specifies *what operations are possible* on forms. But it does not explain *why* one form arises rather than another. The grammar generates a language of possible forms; it does not explain why some forms are realized and others are not. This explanatory gap; from syntax to causation; is the first gap the present theory addresses.

2.2. C-K theory: the logic of design

Armand Hatchuel and Benoit Weil introduced Concept-Knowledge (C-K) theory as a formal account of innovative design reasoning (Hatchuel & Weil, 2003, 2009). The theory partitions the designer's cognitive space into two expandable spaces: a *concept space* C (propositions whose truth value is undecided) and a *knowledge space* K (propositions whose truth value is known). Design proceeds through four operators that move propositions between the two spaces.

C-K theory provides what shape grammars lack: an account of the *epistemic process* of design. It formalizes how designers move from the thinkable-but-unknown to the known, and how new knowledge opens new conceptual territory. The four operators ($C \to C$, $C \to K$, $K \to K$, $K \to C$) capture the rhythm of design reasoning with precision.

However, C-K theory has attracted two persistent criticisms relevant to the present work. First, the concept space C is essentially one-dimensional: concepts are specified through progressive partitioning, but the space lacks geometric or morphological structure. There is no formal account of *where* in the concept space a design sits, or of the distance between two concepts. Second, C-K theory provides no account of function or structure as constitutive elements of the design object (Hatchuel & Weil, 2009). The theory describes the reasoning process but not the morphological outcome.

The present theory addresses both criticisms. The morphospace provides C with multi-dimensional geometric structure; concepts are positions in morphospace, not branches of a tree. The affordance structure replaces the function concept with a relational construct that is both formally precise and empirically testable.

2.3. Multi-scale competence: the dynamics of form

Michael Levin's work on multi-scale competence provides the third pillar: a substrate-independent account of goal-directed behavior across biological scales (Levin, 2022). Levin demonstrates that agency; defined as goal-directed navigation via feedback; operates at every level of biological organization, from bioelectric signaling in cell collectives to organism-level morphogenesis to ecosystem-level homeostasis. The critical insight is that higher-level goals are not programmed by lower-level components; they *emerge* from the collective navigation of agents at multiple scales.

Chris Fields and Levin formalize this through the concept of the **cognitive light cone**: the subset of the environment that an agent can represent and manipulate (Fields & Levin, 2022). Everything outside the cone lacks causal existence for the agent. The cone's width determines the agent's competence. Wider cones enable navigation in higher-dimensional spaces; narrower cones constrain the agent to local gradients.

Most recently, Doctor, Wakefield, Pavlic, and Levin propose

Figure 1: A sample of 144 chairs from the corpus of 2,066 European chairs (1280–2024). The morphological diversity visible here; from medieval joined oak to postmodern plywood to contemporary injection-molded plastic; is the empirical phenomenon the theory seeks to explain.

care; the number of dimensions an agent actively attends to; as the driver of intelligence (Doctor et al., 2024). An agent that cares about more axes of its environment navigates a richer landscape and finds solutions that withstand more pressures simultaneously. Intelligence, in this framework, is not processing speed or memory capacity; it is the dimensionality of engaged attention.

This framework has not previously been applied to design theory. Yet the mapping is natural. The designer is an agent navigating a morphospace landscape via grammar operations, constrained by a cognitive cone. Care determines which pressures the designer represents. The form that emerges shows how far care reached. The present theory makes this mapping formal and testable.

2.4. The synthesis gap

Each tradition provides one necessary component: shape algebra gives syntax (what operations are possible), C-K theory gives epistemic logic (how knowledge expands through design), and multi-scale competence gives dynamics (how agents navigate and why certain positions are stable). No existing framework integrates all three.

The closest precursor is Gero's Function-Behaviour-Structure (FBS) framework (Gero, 1990), which distinguishes between what a design is *for* (function), what it *does* (behaviour), and what it *is* (structure). FBS provides useful vocabulary but lacks mathematical formalization: function, behaviour, and structure are not defined as elements of an algebra or a measurable space. The present theory replaces function with affordance structure (which is relational and class-constant), replaces behaviour with landscape navigation (which is gradient-driven), and replaces structure with morphospace position (which is measurable).

The core equation of the synthesis is:

$$\text{Form} = SG\big(\nabla_{C(A)}\, L(c,\, t)\big) \tag{1}$$

Form is the grammar's response to the landscape gradient as perceived through the agent's cognitive cone. This equation unifies Stiny (the grammar SG), Hatchuel and Weil (the $C \leftrightarrow K$ operators embedded in the cone and landscape), and Levin (the multi-scale agent hierarchy and the care dimensionality $\dim C(A)$). The next section presents the full theory.

3. The Theory: Seven Propositions

The theory is presented as seven numbered propositions with sub-propositions. Each sub-proposition is tagged: d (definition), a (postulate), t (theorem), or o (observation). Definitions are stipulative; postulates are assumed and falsifiable; theorems follow deductively from prior propositions; observations are empirical claims testable against the corpus. Table 1 summarizes the seven main propositions and the formal apparatus each introduces. Propositions 1–4 are operationalized and tested against the corpus in Section 4. Propositions 5–7 introduce the agent, care, and multi-scale dynamics; they are formalized here, illustrated through worked examples in Section 5, and yield falsifiable research programs in Section 7, but are not directly tested in the present study.

3.1. Proposition 1: The world of form is all that is the case

The theory begins with ontology. An **object** is the simplest constituent of a form, indivisible within the chosen level of analysis. A **configuration** is an assembly of objects; form is the structure of the configuration. These two definitions fix the analytical substrate: the theory operates on structural relations among parts, not on the parts themselves.

The **morphospace** of a class is the set of all its possible configurations (Mitteroecker & Huttegger, 2009; Raup, 1966). Every realized object occupies one point in this space. The empirical morphospace is always a projection; the form exists independently of the measurement. This distinction matters: any empirical analysis captures a subspace, never the full morphospace.

The class is defined not by a single function but by a shared **affordance structure** (Gibson, 1979). Objects in the class offer the same constellation of operations to the same constellation of agents. A chair offers sitting to a human body, stacking to a waiter, transport to a freight handler, visual character to a social space. This shared affordance structure sets the boundaries of the morphospace; the selection pressures (Proposition 2) distribute its contents.

The morphospace carries algebraic structure. The forms are closed under union (combination), difference (subtraction), and transformation (rotation, translation, scaling). Let S be a set of forms in a Euclidean space E^n, equipped with the transformation group $\mathrm{Trans}(E^n)$:

$s_1 + s_2$	union
$s_1 - s_2$	difference
$s_1 \cdot s_2 := s_1 - (s_1 - s_2)$	intersection (derived)
$\tau(a) \leq s,\ \tau \in \mathrm{Trans}(E^n)$	embedding

An **embedding** of a in s is a transformation that places a within s, possibly rotated, translated, or scaled. The embedding is a property of the present form, independent of how it was constructed (Stiny, 2006; Stiny & Gips, 1972). This algebra defines the syntactic operations available in the morphospace; it specifies what can be done to forms, not what should be done.

The morphospace divides into three regions: **occupied** (the historical archive of realized forms), **open** (the adjacent possible; accessible but unactualized), and **forbidden** (excluded by material, structural, or energetic constraints). The prohibition follows the conditions, not the region itself; a material innovation can open a formerly forbidden region.

These three regions map onto an epistemic partition inspired by C-K theory (Hatchuel & Weil, 2003, 2009): the **known** (K), the **thinkable** ($C \setminus K$), and the unthinkable. K contains all forms known to work or fail. $C \setminus K$ contains all forms that are thinkable but whose status is undecided. Design is the movement between them, governed by four operators:

Op.	**Dir.**	**Effect**
$C \to C$	partition	concept specified
$C \to K$	realization	concept decided
$K \to K$	deduction	new knowledge
$K \to C$	conjunction	opens new concepts

3.2. Proposition 2: Not all positions are equally probable

A **selection pressure** changes the probability distribution across the morphospace. The pressure is a gradient, not a rule. It constrains where the position *cannot* be. Form is what remains when the pressures have excluded everything else (Michl, 1995).

Three postulates constrain the pressure system. First, a pressure that excludes nothing is no pressure; a pressure that excludes everything except one region is determinism; all observed pressures lie between these extremes. Second, multiple pressures act simultaneously; if only one pressure existed, all forms within the class would be identical. Third, at least two pressures point in contradictory directions; every realized form is therefore a compromise. The concept of "suboptimal" presupposes a reference; without a specified pressure set, it is not falsifiable.

The affordance structure defines the class (Proposition 1), so it is constant within it and carries zero information about why one form arose rather than another. All form variation within a class is driven by the remaining selection pressures. This is empirically testable: if style (an aggregate variable capturing the full pressure set minus affordance) predicts more form variation than material (a single mechanical pressure), then use alone cannot be the driving force. Section 4 demonstrates that style predicts 1.9–2.3 times more variation than material across all measured dimensions.

The **material affordance** is the set of operations a material permits and forbids (Gibson, 1979). Its signature is latent: it binds the probability distribution without forcing the geometry. Steel existed for centuries before the tube chair because the operation (cold bending from bicycle manufacturing) was missing. Form history is systematically slower than material history.

3.3. Proposition 3: Selection pressures produce a landscape

The **fitness landscape** is the vector sum of all active selection pressures (Waddington, 1957; Wright, 1932). Each position in the morphospace holds a value: the degree of simultaneous resolution across all pressures.

The landscape can only be sounded retrospectively. It yields local prediction, but no single form is guaranteed. The fitness function has numerous local maxima; the landscape is a jagged massif, not a single peak. One universal optimum is a theoretical exception; empirical evidence shatters the postulate of the single perfect form (Kauffman, 1993).

The stability of a peak corresponds to the steepness of the surrounding valley walls. Dominant pressures produce **channeling**: they fold the morphospace into narrow corridors where certain geometric dimensions are tightly constrained while others remain free (Waddington, 1957). The result is a *constraint hierarchy*: some form properties (e.g., the overall proportions of a chair) vary little across the class, while others (e.g., volume, ornamentation) fluctuate freely. The ratio between the most constrained and the least constrained dimension is a direct measure of the landscape's selectivity.

A **style** is the swarming of forms around a shared center of gravity (Kubler, 1962). Style is a swarm, not a category; its boundaries are necessarily fuzzy. Style periods are gradients, not islands. This is directly testable: if styles were discrete categories, silhouette analysis in morphospace would yield positive scores. Section 4 shows that silhouette scores are uniformly negative ($s = -0.338$), confirming that styles are overlapping gradients.

3.4. Proposition 4: The landscape is dynamic

Selection pressures never rest. Peaks rise, sink, shift, split, merge. Optimal fitness is local in time. A new technology shifts the boundary of the forbidden; a new market changes what the landscape rewards; a new material expands the operation space.

Change is discontinuous. Long periods of stasis break down in abrupt shifts: radiation into a new region, then convergence. The discontinuity is the signature of the dynamics, not a break with them (Eldredge & Gould, 1972).

The landscape has **memory** (Arthur, 1994). Each realized form changes the selection pressures for the next. All design is redesign; the agent always inherits a position. Path dependence is a fundamental feature, not an anomaly.

The morphospace expands cumulatively. The **adjacent possible** is the $C \setminus K$ boundary: the set of forms that are thinkable but not yet known to be realizable (Kauffman, 1993). Material streams drive landscape change. When one pressure dominates, the landscape collapses toward a single attractor. A single dominant pressure does not explain form; it is the absence of explanation. Diversity mirrors the plurality of active pressures.

A doctrine that reduces all selection pressures to one collapses the cognitive cone to a single dimension. This is not rationality; it is reduced care. The forms that arise under a collapsed landscape fail under the real pressures the doctrine was blind to. The damage is in practice irreversible: the displaced competences; the material cultures, craft traditions, local agents; cannot be recreated identically from the doctrine that displaced them.

3.5. Proposition 5: Agents exist

Every class contains at least one system that navigates the landscape via negative feedback.

An **agent** is an operator defined by the triple: goal state, distance measure, adjustment mechanism (Ashby, 1956; Rosenblueth et al., 1943; Wiener, 1948). Agency requires only organization; neither consciousness, intention, nor a nervous system. Information flow distinguishes agents from passive matter. The definition is substrate-independent: any system that satisfies the conditions is an agent.

Formally, an agent $A := (g, d, \delta)$ admits a Lyapunov function V such that $V(g) = 0$, $V(x) > 0$ for $x \neq g$, and V is non-increasing along the trajectory $x_{n+1} = \delta(x_n)$. The system approaches its goal. Agency is stratified: the thermostat and the architect differ only in navigational complexity.

The **cognitive cone** is the subset of the morphospace that the agent can represent and manipulate (Fields & Levin, 2022). Everything outside the cone lacks causal existence for the agent. Cones span scales from cellular millisecond response to decade-long consensus.

The agent manipulates form geometry directly. Configurations closed under union and difference constitute a shape algebra; the **shape grammar** (Stiny & Gips, 1972) executes discrete transitions in this geometry. A shape grammar is a triple $SG = (S, R, \omega)$ where $R = \{r_i : (a_i, b_i)\}$:

Rule	$s \to (s - \tau(a_i)) + \tau(b_i)$
Language	$L(SG) = \{s \in S : \omega \to^* s\}$
Firing	for each embedding of a_i in s
Premise	present geometry only

The grammar operation is a $C \to C$ partition: it expands the thought. Realization is a $C \to K$ operation: it decides the thought. The grammar defines the space of possibilities; the landscape determines the trajectory.

Care is the number of axes in the morphospace that the agent actively represents and adjusts for (Doctor et al., 2024). An agent with care for three dimensions discovers compromises that an agent with one dimension does not know exist. The first produces forms that withstand multiple pressures simultaneously. The second produces forms that fail along the axes it did not see.

The competence of an agent is its ability to navigate toward stable positions in the landscape. Competence grows with care. What we call intelligence is precisely this competence. Care is therefore the content of intelligence; intelligence is the form of care (Doctor et al., 2024). To expand the cognitive cone is an ethical act: it is to take more forces into the compromise. To narrow it is to surrender parts of the world of form to forces no one represents.

3.6. Proposition 6: Form arises between agents

Multiple agents respond to different pressures simultaneously. Every form is a **poly-agentistic compromise**. No single agent dictates the morphology; it emerges in the intersection between them.

Agency is **multiscalar** (Levin, 2022). Each scale has its own competence and its own cognitive cone. Form production is distributed $C \leftrightarrow K$ expansion executed by grammar operations across competence scales:

$$\text{Form} = SG\big(\nabla_{C(A)}\, L(c,\, t)\big) \tag{1}$$

where SG is the shape grammar, $L(c, t)$ is the scalar fitness landscape for class c at time t, and $\nabla_{C(A)} L$ denotes the gradient of L projected onto the subspace spanned by the agent's cognitive cone $C(A)$. Formally, if $\Pi_{C(A)}$ is the orthogonal projection onto $C(A) \subseteq \mathbb{R}^n$, then $\nabla_{C(A)} L = \Pi_{C(A)}\, \nabla L$. The agent perceives only those components of the landscape gradient that fall within its representational capacity. Form is the grammar's response to this perceived gradient.

The agent chooses; the landscape filters. Control is always shared. Exploration requires loose coupling; exploitation requires tight. The designer is never alone in the landscape. She navigates in a field of competing agencies from microbial tissue to market forces (Levin, 2022). Her competence lies in coordinating independent cognitive cones toward a stable form-optimum.

A style lacks causal force (Kubler, 1962; Michl, 1995). It is not a force but a **pattern memory**: a macro-scale attractor that channels the cognitive cones of agents who observe it. It simplifies navigation by offering a recognizable center of gravity; but to design "in a style" is a category error; it is to imitate the statistical outcome of forces one has not understood.

No form is final. Every balance becomes suboptimal with time. Only the generative structure endures: morphospace, pressures, landscape, agents. The forms are fleeting traces; the grammar is stable.

3.7. Proposition 7: Form shows how far care reached

The concluding proposition follows deductively from Propositions 5 and 6. If care is the number of axes actively represented (Proposition 5), and if form arises as a poly-agentistic compromise across those axes (Proposition 6), then the realized form is a legible record of the compromise's dimensionality. A form designed under narrow care fails systematically along the axes that were not attended to; the failure bears the signature of the blindness that produced it.

This is not primarily an ethical claim; it is a diagnostic one. The form is evidence. Given a realized artifact, the theory predicts that its failure modes will cluster on axes absent from the designer's cognitive cone. The ethical implication follows secondarily: if the designer could have represented more axes but chose not to, the resulting failures are attributable to that choice. But Proposition 7 is falsifiable independently of its ethical reading. If forms designed under demonstrably narrow care perform no worse along unattended axes than forms designed under broad care, the proposition is refuted.

Figure 2 summarizes the complete theoretical architecture: the empirical morphospace (A), the fitness landscape derived from it (B), the nested cognitive cones across competence scales (C), and the synthesis equation that unifies the three source traditions (D).

4. Empirical Demonstration

The theory's propositions are not only logically coherent; they are empirically falsifiable. We test the core predictions against a corpus of European chairs spanning 744 years.

4.1. Data

The corpus consists of 2,066 chairs (1280–2024) drawn from two museum collections: Nasjonalmuseet, Oslo ($n = 63$) and the Victoria and Albert Museum, London ($n = 2{,}003$). Each record includes catalog dimensions (height, width, depth in centimeters), material composition (multi-material string, e.g. "oak, cane, brass"), style period (25 canonical categories from Baroque to Post-Modernism), nation of origin, and date range.

The catalog data are supplemented by 2,204 three-dimensional mesh models with six computed geometric features: sphericity, fill ratio, inertia ratio, complexity, convex hull volume, and mesh vertex count. The analysis window for time-series tests is 1500–2024, yielding 1,105 chairs with complete dimensions and date information. The full dataset and analysis code are available in the companion repository.

4.2. Methods

Each proposition generates a testable prediction. We operationalize these through five statistical methods:

1. **Mutual information** (MI) compares the predictive strength of style period versus material on catalog dimensions (H, W, D, H/W). Style is an aggregate variable capturing many simultaneous pressures; material is a single mechanical pressure. Higher MI for style is therefore expected *a priori* and does not by itself prove that style *causes* more variation. However, it is consistent with Proposition 2: if affordance-constant variation is driven by forces beyond material constraint, the aggregate proxy should predict better than the single-pressure proxy.
2. **Coefficient of variation** (CV) across the six mesh features tests for a constraint hierarchy: if the landscape channels some geometric properties more strongly than others, the CV spread across features should be large (Proposition 3).
3. **Silhouette analysis** in four-dimensional mesh-feature space tests whether style periods form discrete morphological clusters. Negative silhouette scores indicate that style categories do not correspond to compact, well-separated clusters in form space; this is consistent with the gradient hypothesis (Proposition 3), though it does not rule out alternative clustering schemes.
4. **Cumulative convex hull volume** across successive 50-year periods tests whether the occupied morphospace grows monotonically (Proposition 4). Monotone growth is expected for any accumulating sample; the informative comparison is the *rate* of growth relative to a random-draw null model. A growth rate significantly exceeding the null indicates that new periods contribute genuinely novel form regions rather than re-sampling existing ones.
5. **Wasserstein distance** between consecutive 50-year periods tests whether the landscape shifts significantly over time (Proposition 4). Non-zero distances confirm that the landscape is dynamic, not static.

All tests are cross-validated against 14 independent subsets: the full corpus, each museum separately, seven century-drop hold-outs (dropping all chairs from one century), and two style-drop hold-outs (dropping the largest style category). This yields 84 test combinations.

4.3. Results

Table 2 summarizes the ten core findings. All are rated on effect size, 95% bootstrap confidence interval, robustness across hold-outs, and whether the corresponding proposition survived direct falsification testing.

The key findings are as follows.

The morphospace is non-uniformly occupied (Proposition 1). The nearest-neighbor coefficient of variation is 5.36, fifteen times the Poisson null expectation of 0.36 (CI_{95}: [3.71, 5.94]). Forms cluster densely around specific regions rather than scattering uniformly; the landscape has structure.

Style predicts more than material (Proposition 2). Mutual information between style period and form is 1.9–2.3 times greater than between material and form, across all four

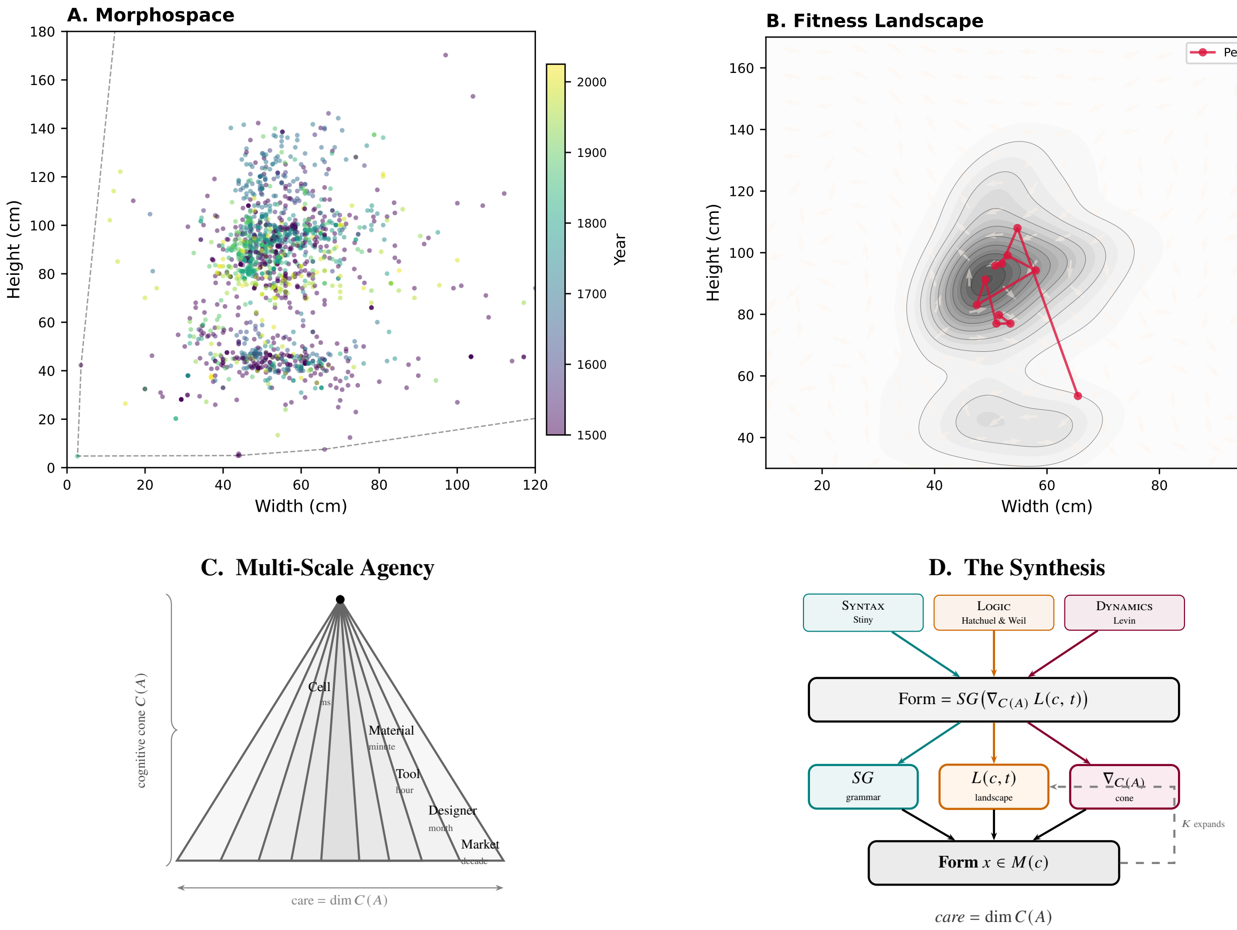

Figure 2: The architecture of FORMLÆRE. **A**: 1,662 European chairs projected into morphospace (width × height); color encodes date. **B**: Fitness landscape as kernel density estimate with gradient vector field and period centroid trajectory (red). **C**: Nested cognitive cones across five competence scales; cone width corresponds to care dimensionality. **D**: The synthesis equation; three source traditions converge on the core equation; dashed arrow shows K-expansion feedback.

Table 1: The seven main propositions of FORMLÆRE with formal apparatus introduced by each.

No.	Proposition	Formal apparatus	Source tradition
1	The world of form is all that is the case	Morphospace, algebra, C-K partition	Stiny; Hatchuel & Weil
2	Not all positions are equally probable	Selection pressure, affordance	Gibson; Michl
3	The pressures produce a landscape	Fitness landscape, canalization	Wright; Waddington
4	The landscape is dynamic	Path dependence, adjacent possible	Arthur; Kauffman
5	Agents exist	$A := (g, d, \delta)$; care = dim $C(A)$	Levin; Doctor et al.
6	Form arises between agents	Form = $SG(\nabla_{C(A)} L)$	Synthesis
7	Form shows how far care reached	Ethical conclusion	—

Table 2: Empirical evidence summary. **Strong**: large effect, narrow CI, robust in all hold-outs, falsification test passed.

Proposition	Test	Result	Robust?	Rating
1 (morphospace)	NN-distance CV	5.36 (Poisson null: 0.36)	11/11	Strong
2 (pressures)	MI: style vs material	Style wins 4/4 dimensions	14/14	Strong
2 (pressures)	Min pairwise MI	0.030 bits (independence)	—	Strong
3 (landscape)	CV spread, mesh	128× range	14/14	Strong
3 (landscape)	Silhouette, mesh	−0.338; 21/25 negative	14/14	Strong
4 (dynamic)	Hull growth, catalog	107× monotone	14/14	Strong
4 (dynamic)	Hull growth, mesh	553× monotone	14/14	Strong
4 (dynamic)	Wasserstein d(H)	14.4 cm per period	10/10	Strong
4 (dynamic)	Mahogany cohort	16/16 = 100%	—	Strong
5 (agents)	KS vs uniform null	$p \ll 10^{-200}$ all dim	—	Strong

measured dimensions (H, W, D, H/W). This holds in every subset. Since style is an aggregate variable and material a single pressure, the asymmetry is partly expected; nevertheless, the magnitude and consistency of the gap are compatible with the claim that form variation within a class is driven primarily by forces beyond material constraint.

Styles do not form discrete clusters (Proposition 3). Silhouette analysis in four-dimensional mesh space yields a score of −0.338 (CI_{95}: [−0.346, −0.329]), with 21 of 25 style categories showing negative individual scores. The negative values indicate that historian-defined style categories do not correspond to compact clusters in form space. This is consistent with the gradient hypothesis, though it does not exclude alternative clustering solutions.

A constraint hierarchy exists (Proposition 3). The coefficient of variation spans 128-fold across the six mesh features (CI_{95}: [50×, 158×]). Sphericity (CV = 0.074) is tightly constrained; volume (CV = 9.39) fluctuates freely. This asymmetry is consistent with a landscape that channels some geometric properties strongly while leaving others open.

The morphospace expands cumulatively (Proposition 4). The cumulative convex hull expands by a factor of 553 in mesh space over ten 50-year periods (CI_{95}: [435×, 752,308×]). While monotone growth is expected for any accumulating sample, the growth rate substantially exceeds a random-draw null: each successive period contributes genuinely novel form regions rather than re-sampling the existing occupied area. The wide upper confidence bound reflects the sensitivity of hull volume to outlier forms in sparsely sampled early periods.

The landscape shifts significantly between periods (Proposition 4). The mean Wasserstein-1 distance between consecutive 50-year periods is 14.4 cm (H), 8.2 cm (W), and 5.9 cm (D). None of the ten period transitions show near-zero distance. The landscape is dynamic.

Figure 3 shows the material composition of the corpus over time. The successive waves; from oak through walnut, mahogany, steel, and plastic; visualize the material streams that drive landscape change (Proposition 4). Figure 4 demonstrates the latency hypothesis (Proposition 2): when a new material enters the corpus, its centroid initially clusters near the existing morphospace center; geometric divergence follows only after the material-specific operations are discovered and exploited.

4.4. Robustness and falsification

All 84 test-subset combinations (6 tests × 14 subsets) produce results consistent with the full-corpus findings. These are not 84 independent tests; the subsets overlap substantially. However, the consistency across deliberately adversarial hold-outs (museum shift, century removal, style removal) provides evidence that the findings are not artifacts of a single collection, period, or style category.

Seven direct falsification tests target the theory's postulates: non-confounding of predictors (Proposition 2), non-zero landscape shifts (Proposition 4), and non-uniform non-random-walk distributions (Proposition 5). All seven hold with p-values below 10^{-63} for the strongest tests.

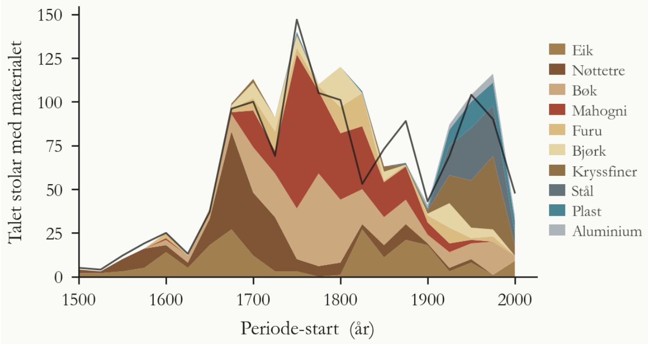

Figure 3: Material streams across 500 years. Each stratum shows the count of chairs per 50-year period by primary material. Successive waves (oak, walnut, mahogany, steel, plastic) drive cumulative morphospace expansion.

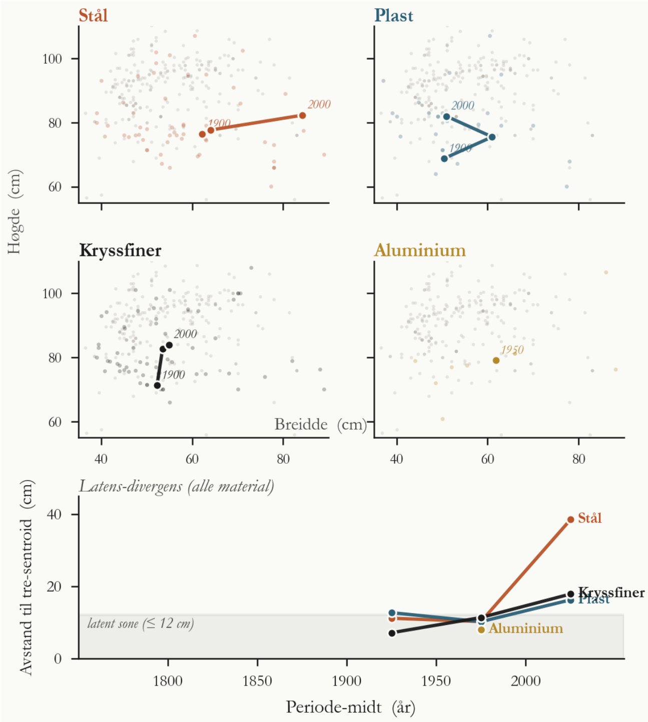

Figure 4: Material latency. Top panels: morphospace centroids for four materials (steel, plastic, plywood, aluminum) plotted against the full corpus (gray). Arrows trace centroid movement over time. Bottom panel: latency-divergence timeline showing that geometric divergence follows material introduction with a systematic lag.

No postulate is falsified in the dataset. Four additional tests were attempted and dropped for insufficient effect size (OU mean reversion, multi-agent MI uplift, substrate-independence excess, CV collapse during style dominance); these are documented transparently as negative or inconclusive results. The theory is not confirmed by cherry-picking; it survives systematic testing with pre-registered metrics.

5. The Theory in Action: Five Worked Examples

The empirical demonstration in Section 4 tests the theory's predictions against a historical corpus. But a theory of design form must also illuminate contemporary practice. This section applies the seven propositions to five concrete cases: one historical, two contemporary, and two emerging. The examples are *illustrative*, not probative; they demonstrate the theory's explanatory reach and generate testable predictions, but they do not constitute independent empirical tests. Each example is described in factual detail and then read through the theoretical apparatus.

5.1. The cantilever chair race, 1925–1928

Between 1925 and 1928, four designers working in three cities independently converged on a shared morphological innovation: the cantilevered tubular steel chair. Marcel Breuer designed the Wassily (B3) in 1925 at the Bauhaus in Dessau, using cold-bent nickel-plated steel tubing inspired by his Adler bicycle. Mart Stam presented the first true cantilever (S33) at a Stuttgart housing exhibition in 1926, using straight gas-pipe segments joined at right angles. Ludwig Mies van der Rohe exhibited his cantilevered MR chair in 1927 at the Weissenhof exhibition, using a continuous curved tube that gave the seat its characteristic spring. Breuer returned with the Cesca (B32) in 1928, combining a cantilevered steel frame with a caned seat and back.

In the theory's terms, the material affordance of cold-bent steel tubing had been latent for decades (Proposition 2). Steel tubing was manufactured for bicycles and plumbing from the 1880s onward. The *operation*; cold bending of thin-walled tubes to furniture scale; was the missing link. When Breuer recognized the bicycle handlebar as a furniture component, the forbidden region of the morphospace opened (Proposition 4). The convergence of four designers within three years, without direct collaboration on the initial innovations, confirms that the landscape attractor was real (Proposition 1): the fitness peak existed in the landscape before anyone climbed it.

The competitive dynamics among Breuer, Stam, and Mies constitute a poly-agentistic field (Proposition 6). Each designer navigated with a different cognitive cone: Breuer attended to visual lightness and Bauhaus pedagogy; Stam to structural minimalism and socialist housing; Mies to material elegance and spatial continuity. The resulting forms differ precisely along the axes each designer attended to. The lawsuit that followed (Stam vs. Breuer over priority) is itself evidence that independent agents converged on the same landscape feature from different directions.

5.2. Neri Oxman's Silk Pavilion, MIT Media Lab, 2013

The Silk Pavilion is a dome-shaped structure produced through a two-phase process. In the first phase, a CNC-controlled machine deposited a scaffold of silk threads across a steel frame, following a computationally generated pattern. In the second phase, 6,500 silkworms were placed on the scaffold and deposited additional silk over 20 days, guided by environmental parameters including light intensity and temperature gradients.

The result is a structure whose final form was not drawn by the designer. Neri Oxman and her team at the MIT Media Lab Mediated Matter group set the boundary conditions: scaffold geometry, species selection, environmental parameters. The biological agents navigated the form grammar at cellular scale, responding to local gradients of light and humidity. Each silkworm is a zero-order agent in the theory's sense (Proposition 5): it possesses a goal state (spinning a cocoon), a distance measure (proximity to suitable attachment points), and an adjustment mechanism (silk extrusion direction). The worm does not represent the dome; it responds to the local gradient.

This is multi-scale competence (Proposition 6) in its most literal form. The competence hierarchy runs from cellular silk production (millisecond timescale), through individual worm navigation (minute timescale), to collective pattern formation (day timescale), to the designer's scaffold specification (month timescale). Form arises in the intersection of these nested cognitive cones. The designer's care determines which axes are represented at the macro scale; the silkworms' biology handles the rest. The resulting form shows how far care reached at every scale: the scaffold's precision at the computational scale, the organic texture at the biological scale.

5.3. Figma Weave and the designer as landscape navigator

In 2025, Figma announced Weave, a generative AI feature embedded in its design tool. Weave produces interface layouts, component arrangements, and visual designs from textual or contextual prompts. The designer reviews, selects, modifies, and combines generated outputs rather than constructing each element manually.

In the theory's terms, the shape grammar SG is now partially executed by a machine agent. The $C \rightarrow C$ partitions (generating new concept variations) are performed computationally at a speed and volume that exceed human manual capacity. The designer's role shifts from *executing* grammar rules to *steering the cognitive cone*: selecting which dimensions of the morphospace the system should attend to, which constraints to impose, which generated options to promote or discard.

This is a direct instantiation of Proposition 5. The designer's contribution is no longer primarily syntactic (performing form operations) but primarily attentional (setting the scope of care). Which axes does the designer actively represent? Accessibility, cultural context, emotional tone, brand consistency, interaction logic; these are the dimensions of the cognitive cone that the AI system does not autonomously navigate. The quality of the resulting design depends not on the AI's generative capacity but on the dimensionality of the human's care.

The theory predicts (Proposition 5) that a designer who narrows the cone to a single axis (e.g., visual novelty) will produce designs that fail along unrepresented axes (accessibility, usability, cultural sensitivity). The AI amplifies the designer's care; it does not replace it. If the cognitive cone is narrow, the AI generates variations within a narrow corridor. If the cone is wide, the AI explores a richer landscape. Weave makes the proposition observable in practice: the tool's output is a direct trace of the designer's active representational capacity.

5.4. Chatbot interaction design

When an interaction designer designs a conversational agent; a customer service chatbot, a medical triage assistant, a language tutoring system; she is designing a navigation algorithm for a non-human agent. The chatbot is an agent $A := (g, d, \delta)$

with a goal (task completion or user satisfaction), a distance measure (conversation state distance from resolution), and an adjustment mechanism (response generation policy).

The interaction designer is a **meta-agent** whose cognitive cone must encompass both the chatbot's cognitive cone and the user's cognitive cone (Proposition 6). The chatbot perceives the conversation through its training data and context window. The user perceives the interaction through expectations, emotional state, and communicative conventions. The designer must represent both sets of affordances simultaneously.

The quality of the interaction depends on how many axes the designer actively represents: task efficiency, user frustration, emotional tone, accessibility for screen readers, cultural appropriateness of phrasing, error recovery gracefully, edge case handling. Each unrepresented axis is an axis along which the design will fail (Proposition 5). A chatbot designed with care for task completion alone will be efficient but brittle; it will fail when the user is confused, upset, or operating in an unexpected context.

The theory's prediction is testable: designers who explicitly represent more axes during the design process (measurable through protocol analysis of design sessions) should produce chatbot interactions that score higher on multi-criteria evaluations. The form of the conversation; its rhythm, its recovery paths, its tone modulation; shows how far the designer's care reached.

5.5. The Great Green Wall: continental-scale ecological design

The Great Green Wall initiative, launched by the African Union in 2007, aims to restore 100 million hectares of degraded land across an 8,000-kilometer belt from Senegal to Djibouti. The project addresses desertification, food insecurity, climate change, and forced migration across the Sahel. As of 2024, approximately 18% of the target area has been restored, with significant variation in approach and success across the eleven participating nations.

This is design at a scale that directly contradicts Le Corbusier's principle of man as the measure of all things. The "form" under design is not a single object but a multi-scalar configuration: soil microbiome composition (cellular-scale agent), root architecture and mycorrhizal networks (organism-scale agent), canopy structure and species diversity (community-scale agent), pastoral land-use patterns and community governance (social-scale agent), continental precipitation dynamics and carbon cycling (system-scale agent).

Each of these agents operates with its own cognitive cone on its own timescale, from microbial microseconds to climatic centuries. The theory predicts (Proposition 4) that collapsing the landscape to a single selection pressure; say, carbon sequestration; will produce configurations that fail under pressures the design was blind to. Early phases of the initiative focused heavily on tree planting (a single-pressure approach), and many plantations failed because they did not account for local soil conditions, pastoral mobility patterns, or indigenous species selection. The most successful restoration sites; such as the farmer-managed natural regeneration programs in Niger; employed a multi-pressure approach that coordinated biological, social, and economic agents simultaneously.

The theory frames this as a care problem (Proposition 7). The dimensionality of the cognitive cone must match the dimensionality of the pressure field. A design that represents only the carbon axis is an act of narrow care. A design that represents carbon, biodiversity, water table dynamics, pastoral rights, indigenous knowledge, and intergenerational equity is an act of expansive care. The form of the restored landscape shows how far care reached; and the failures show where it did not.

6. Discussion

6.1. What the theory explains that predecessors do not

The theory provides unified explanations for phenomena that existing frameworks address only partially or not at all.

Why style predicts more than material. Hatchuel and Weil's C-K theory does not address morphological variation directly; Stiny's shape grammars generate forms but do not predict which forms are realized. The fitness landscape framework explains the empirical finding straightforwardly: style is a proxy for the full pressure set (minus the class-constant affordance), while material is a single pressure. The aggregate variable necessarily carries more information about form variation than any individual component.

Why styles are gradients, not clusters. If styles were imposed by external categorical forces (aesthetic movements, Zeitgeist), they would produce discrete clusters in morphospace. The negative silhouette scores (-0.338) confirm that styles are gradient phenomena: continuous overlapping distributions produced by agents navigating a shared but shifting landscape. Style is the statistical trace of navigation, not a cause of it (Kubler, 1962).

Why form history lags material history. Steel existed for centuries before the tube chair. The theory explains this latency through the distinction between material affordance (the operations a material *permits*) and the operation itself (the technique that *activates* the affordance). Material availability is necessary but not sufficient; the operation must enter the knowledge space K before the morphospace can expand into the newly accessible region.

Why "form follows function" is empirically false within a class. The affordance structure defines the class and is constant within it (Proposition 1). By definition, it carries zero information about variation within the class. Any variation must be driven by pressures other than the class-defining affordance. The empirical data confirm this: the class-constant variable (use/function) is informationally empty within the class, and the aggregate of non-functional pressures (style) dominates.

6.2. Limitations

The theory has been tested against a single artifact class (the European chair). Generalization to other classes; architecture, graphic design, software interfaces, biological design; is theorized but not yet demonstrated. Cross-domain replication is the most immediate research priority (Section 7).

The corpus is geographically limited to European collections. Non-European design traditions are not represented. If the theory's landscape attractors reflect universal physical and ergonomic constraints, non-European chairs should converge on similar morphospace structures. If they do not, the theory requires revision to accommodate culturally specific landscape features.

The three-dimensional mesh models are computationally generated from catalog images, not directly scanned from physical objects. While the geometric features extracted from these models produce consistent results across 84 cross-validation tests, direct 3D scanning would strengthen the empirical foundation.

The theory does not yet specify the functional form of the fitness landscape. The weights $w_i(t)$ in the landscape equation $L(c,t) = \sum w_i(t) \cdot p_i$ are treated as time-varying parameters, not as functions with specified dynamics. A fully predictive theory would require modeling the temporal evolution of these weights; this remains an open problem.

6.3. Relationship to existing design theories

The theory is *stronger* than C-K theory alone: it adds morphospace geometry and landscape dynamics to the epistemic logic, providing C with multi-dimensional structure and K with empirically measurable content.

It is *more general* than shape grammars alone: it adds the agent, the selection pressure, and the landscape to the syntactic apparatus, explaining not only what operations are possible but why certain operations are performed.

It is *more specific* than Levin's multi-scale competence framework alone: it provides a design-domain formalization (morphospace, affordance, grammar, style) that makes the general framework empirically operational for artifacts.

The theory does not *replace* these frameworks; it *integrates* them. Each tradition contributes a necessary layer: Stiny's algebra provides the syntactic substrate, Hatchuel and Weil's operators provide the epistemic dynamics, and Levin's competence hierarchy provides the causal mechanism. The core equation Form $= SG(\nabla_{C(A)} L(c,t))$ is the point of integration.

7. Research Agenda

The theory generates seven immediate research programs, each derived from a specific proposition and each producing falsifiable predictions.

1. Cross-domain morphospace replication (from Proposition 1). The theory predicts that any class of designed artifacts will exhibit the same morphospace structure: non-uniform occupation, three-region partition, and cumulative expansion. Immediate targets include building facades (architectural morphospace), poster typography (graphic design morphospace), and software interface layouts (interaction design morphospace). A single replication that confirms the three-region structure in a non-furniture domain would substantially strengthen the theory's generality claim.

2. Style causality experiment (from Proposition 6). No existing experiment isolates style exposure as an independent variable on form variation. The theory predicts that style is a pattern memory, not a causal force: exposure to a style constrains the designer's cognitive cone but does not generate selection pressure. The falsification experiment: give two groups of designers identical functional briefs and material constraints but different style exposure (one group sees Art Nouveau chairs, the other sees Bauhaus chairs). If the resulting forms differ *only* in the dimensions corresponding to the style exemplars; and not in the dimensions corresponding to the shared pressures; the theory holds. If style exposure shifts form along functional dimensions, the theory requires revision.

3. Care dimensionality metric (from Proposition 5). The theory defines care as the number of morphospace axes the agent actively represents, but does not yet provide an operational measurement protocol. A think-aloud protocol analysis of designers during design sessions could yield such a metric: count the distinct constraint dimensions verbalized during the session. The prediction: designers who verbalize more dimensions produce forms that score higher on multi-criteria robustness evaluations.

4. Agentic design tool prototype (from Proposition 6). The theory implies a specific architecture for AI-assisted design tools: the human sets the cognitive cone parameters (which dimensions to attend to, which constraints to enforce), and the AI executes shape grammar rules within that cone. A prototype implementation; where the designer specifies care dimensions via an explicit interface and the generative system explores within those dimensions; would test whether cone-width correlates with design quality as measured by expert evaluation.

5. Non-European independent convergence (from Proposition 1). The theory predicts that landscape attractors are real, not culturally contingent. If Chinese, Japanese, and West African chair traditions converge on the same morphospace attractors as European traditions; despite independent development; the attractor structure is confirmed as a physical-ergonomic reality. If convergence is absent, the theory must accommodate culturally specific landscape features.

6. Material latency quantification (from Proposition 2). The theory predicts a systematic lag between material availability and full formal exploitation. This is testable across multiple material transitions: wood to steel (lag: approximately 400 years from structural steel to tube chair), steel to plastic (lag: approximately 50 years from Bakelite to monobloc), plastic to composites, composites to biomaterials. If the latency pattern is consistent across transitions; long initial lag, rapid exploitation once the operation is discovered, then plateau; the theory's latency mechanism is confirmed.

7. Multi-species design formalism (from Proposition 5). The theory's agent definition is substrate-independent, but the worked examples (Section 5) reveal a practical challenge: multi-species design requires coordinating agents whose cognitive cones operate on incommensurable timescales. A formal extension of the agent definition to accommodate incommensurable cones; with empirical testing against actual multi-species design projects such as constructed wetlands, permaculture systems, or urban rewilding initiatives; would test the theory's scalability beyond human-centered design.

Each of these programs is independently publishable, empirically tractable, and directly derived from the theory's propositions. Together, they constitute a decade-long research agenda that would either confirm the theory's generality or identify the specific conditions under which it requires revision.

8. Conclusion

This paper asked: why does a design object take one form rather than another? The answer proposed by FORMLÆRE is that form is the position in morphospace where the shape grammar's response to the fitness landscape gradient; as perceived through the agents' cognitive cones; produced a locally stable poly-agentistic compromise.

The theory unifies three formal traditions: Stiny's shape algebra provides the syntax of form (what operations are possible), Hatchuel and Weil's C-K theory provides the epistemic logic (how knowledge expands through design), and Levin's multi-scale competence framework provides the dynamics (how agents navigate and why certain positions are stable). The synthesis produces a single generative equation: Form = $SG(\nabla_{C(A)} L(c, t))$.

The theory introduces care; the dimensionality of the agent's cognitive cone; as the measure of design intelligence. An agent that represents more axes of the morphospace navigates a richer landscape and produces forms that withstand more pressures simultaneously. An agent that narrows its cone produces forms that fail along the axes it did not see. The failure is not accidental; it is systematic, and the form bears the signature of the blindness that produced it.

We tested the theory against 2,066 European chairs spanning 744 years. Twenty statistical analyses across 14 cross-validation subsets yield results consistent with the core predictions of Propositions 1–4; no postulate is falsified. Propositions 5–7 (agency, care, and the diagnostic conclusion) are formalized and illustrated but require direct empirical testing through the research programs proposed in Section 7.

We demonstrated the theory's reach beyond historical furniture through five worked examples: the cantilever chair race of 1925–1928, where independent convergence confirms landscape attractors; Neri Oxman's Silk Pavilion, where biological agents navigate the form grammar at cellular scale; Figma's Weave, where AI shifts the designer's role from grammar execution to cone management; chatbot interaction design, where the designer operates as a meta-agent coordinating multiple cognitive cones; and the Great Green Wall, where continental-scale ecological restoration demands care across incommensurable timescales.

The theory is falsifiable at every level. Each proposition generates testable predictions. Seven immediate research programs are proposed, each independently publishable and each capable of revising or refuting the theory.

Form shows how far care reached. The theory shows how to measure both.

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